8.2 Overloading

GPRS as a technology means a newer and faster way for transferring data and as with all networks, data transfer consumes bandwidth and clogs the network. Voice and GPRS data transfers are bound to affect network performance. Similar to the "Network Not Available" message that you may sometimes receive when trying to make a call at peak hours, data transfer speeds are also affected during peak hours. While GPRS and voice channels use the same network resources, the GPRS technology is intelligent to manage or allocate the channels dynamically allowing a reduction in peak time signaling channel loading by sending short messages over GPRS channels instead.

Lower Speeds

The theoretical speed of the GPRS data transmission speed is 172.2 kbps (dependent on the networks and terminals used) but this is only possible in a scenario when one single user without any error protection is hogging all the timeslots. Any service provider will obviously not allow this and in all probability you will be able to access only one or two time slots that will limit the amount of bandwidth available to you. This translates into lower data transmission speeds.

No Store And Forward Mechanism

The main advantage of SMS is the store and forward mechanism. What this essentially means is that any SMS that you send using your mobile first goes to a central server and is then forwarded to the destination phone number. In case that number is switched off or unavailable, the message is stored on the server for some amount of time. This is absent in the GPRS standard.

Data Transit Speeds

GPRS data transfer takes place using a simple principle. Data sent from the sender is transmitted to the recipient using different paths. The probability of the data being lost or corrupted during transit is high and hence the GPRS standard includes data integrity and retransmission strategies. Nevertheless, due to this, poten-